iteration rather than as a separate training phase, but found that in the early phase of training, the coverage objective interfered with the main objective, reducing overall performance.

6 Results

6.1 Preliminaries

Our results are given in Table 1. We evaluate our models with the standard ROUGE metric (Lin, 2004b), reporting the F_1 scores for ROUGE-1, ROUGE-2 and ROUGE-L (which respectively measure the word-overlap, bigram-overlap, and longest common sequence between the reference summary and the summary to be evaluated). We obtain our ROUGE scores using the pyrouge package.[4] We also evaluate with the METEOR metric (Denkowski and Lavie, 2014), both in exact match mode (rewarding only exact matches between words) and full mode (which additionally rewards matching stems, synonyms and paraphrases).[5]

In addition to our own models, we also report the lead-3 baseline (which uses the first three sentences of the article as a summary), and compare to the only existing abstractive (Nallapati et al., 2016) and extractive (Nallapati et al., 2017) models on the full dataset. The output of our models is available online.[6]

Given that we generate plain-text summaries but Nallapati et al. (2016; 2017) generate anonymized summaries (see Section 4), our ROUGE scores are not strictly comparable. There is evidence to suggest that the original-text dataset may result in higher ROUGE scores in general than the anonymized dataset – the lead-3 baseline is higher on the former than the latter. One possible explanation is that multi-word named entities lead to a higher rate of n-gram overlap. Unfortunately, ROUGE is the only available means of comparison with Nallapati et al.'s work. Nevertheless, given that the disparity in the lead-3 scores is (+1.1 ROUGE-1, +2.0 ROUGE-2, +1.1 ROUGE-L) points respectively, and our best model scores exceed Nallapati et al. (2016) by (+4.07 ROUGE-1, +3.98 ROUGE-2, +3.73 ROUGE-L) points, we may estimate that we outperform the only previous abstractive system by at least 2 ROUGE points all-round.

[4] pypi.python.org/pypi/pyrouge/0.1.3
[5] www.cs.cmu.edu/~alavie/METEOR
[6] www.github.com/abisee/pointer-generator

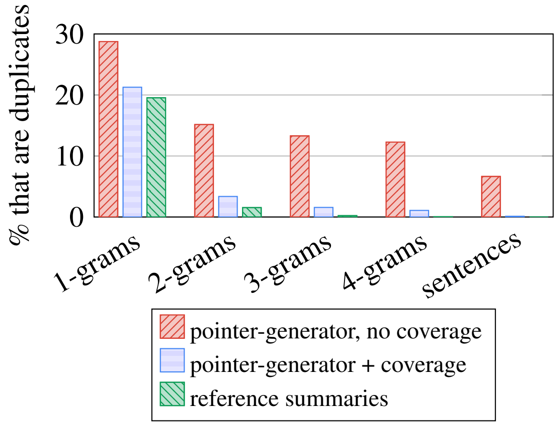

Figure 4: Coverage eliminates undesirable repetition. Summaries from our **non-coverage model** contain many duplicated n-grams while our **coverage model** produces a similar number as the **reference summaries**.

6.2 Observations

We find that both our baseline models perform poorly with respect to ROUGE and METEOR, and in fact the larger vocabulary size (150k) does not seem to help. Even the better-performing baseline (with 50k vocabulary) produces summaries with several common problems. Factual details are frequently reproduced incorrectly, often replacing an uncommon (but in-vocabulary) word with a more-common alternative. For example in Figure 1, the baseline model appears to struggle with the rare word *thwart*, producing *destabilize* instead, which leads to the fabricated phrase *destabilize nigeria's economy*. Even more catastrophically, the summaries sometimes devolve into repetitive nonsense, such as the third sentence produced by the baseline model in Figure 1. In addition, the baseline model can't reproduce out-of-vocabulary words (such as *muhammadu buhari* in Figure 1). Further examples of all these problems are provided in the supplementary material.

Our pointer-generator model achieves much better ROUGE and METEOR scores than the baseline, despite many fewer training epochs. The difference in the summaries is also marked: out-of-vocabulary words are handled easily, factual details are almost always copied correctly, and there are no fabrications (see Figure 1). However, repetition is still very common.

Our pointer-generator model with coverage improves the ROUGE and METEOR scores further, convincingly surpassing the best abstractive model

Article: smugglers lure arab and african migrants by offering discounts to get onto overcrowded ships if people bring more potential passengers, a cnn investigation has revealed. (...)
Summary: cnn investigation **uncovers** the **business inside** a **human smuggling ring**.

Article: eyewitness video showing white north charleston police officer michael slager shooting to death an unarmed black man has exposed discrepancies in the reports of the first officers on the scene. (...)
Summary: more **questions than answers emerge** in **controversial s.c.** police shooting.

Figure 5: Examples of highly abstractive reference summaries (**bold** denotes novel words).

of Nallapati et al. (2016) by several ROUGE points. Despite the brevity of the coverage training phase (about 1% of the total training time), the repetition problem is almost completely eliminated, which can be seen both qualitatively (Figure 1) and quantitatively (Figure 4). However, our best model does not quite surpass the ROUGE scores of the lead-3 baseline, nor the current best extractive model (Nallapati et al., 2017). We discuss this issue in section 7.1.

7 Discussion

7.1 Comparison with extractive systems

It is clear from Table 1 that extractive systems tend to achieve higher ROUGE scores than abstractive, and that the extractive lead-3 baseline is extremely strong (even the best extractive system beats it by only a small margin). We offer two possible explanations for these observations.

Firstly, news articles tend to be structured with the most important information at the start; this partially explains the strength of the lead-3 baseline. Indeed, we found that using only the first 400 tokens (about 20 sentences) of the article yielded significantly higher ROUGE scores than using the first 800 tokens.

Secondly, the nature of the task and the ROUGE metric make extractive approaches and the lead-3 baseline difficult to beat. The choice of content for the reference summaries is quite subjective – sometimes the sentences form a self-contained summary; other times they simply showcase a few interesting details from the article. Given that the articles contain 39 sentences on average, there are many equally valid ways to choose 3 or 4 highlights in this style. Abstraction introduces even more options (choice of phrasing), further decreasing the likelihood of matching the reference summary. For example, *smugglers profit from desperate migrants* is a valid alternative abstractive summary for the first example in Figure 5, but it scores 0 ROUGE with respect to the reference summary. This inflexibility of ROUGE is exacerbated by only having one reference summary, which has been shown to lower ROUGE's reliability compared to multiple reference summaries (Lin, 2004a).

Due to the subjectivity of the task and thus the diversity of valid summaries, it seems that ROUGE rewards safe strategies such as selecting the first-appearing content, or preserving original phrasing. While the reference summaries *do* sometimes deviate from these techniques, those deviations are unpredictable enough that the safer strategy obtains higher ROUGE scores on average. This may explain why extractive systems tend to obtain higher ROUGE scores than abstractive, and even extractive systems do not significantly exceed the lead-3 baseline.

To explore this issue further, we evaluated our systems with the METEOR metric, which rewards not only exact word matches, but also matching stems, synonyms and paraphrases (from a predefined list). We observe that all our models receive over 1 METEOR point boost by the inclusion of stem, synonym and paraphrase matching, indicating that they may be performing some abstraction. However, we again observe that the lead-3 baseline is not surpassed by our models. It may be that news article style makes the lead-3 baseline very strong with respect to any metric. We believe that investigating this issue further is an important direction for future work.

7.2 How abstractive is our model?

We have shown that our pointer mechanism makes our abstractive system more reliable, copying factual details correctly more often. But does the ease of copying make our system any less *abstractive*?

Figure 6 shows that our final model's summaries contain a much lower rate of novel n-grams (i.e., those that don't appear in the article) than the reference summaries, indicating a lower degree of abstraction. Note that the baseline model produces novel n-grams more frequently – however, this statistic includes all the incorrectly copied words, *UNK* tokens and fabrications alongside the good instances of abstraction.

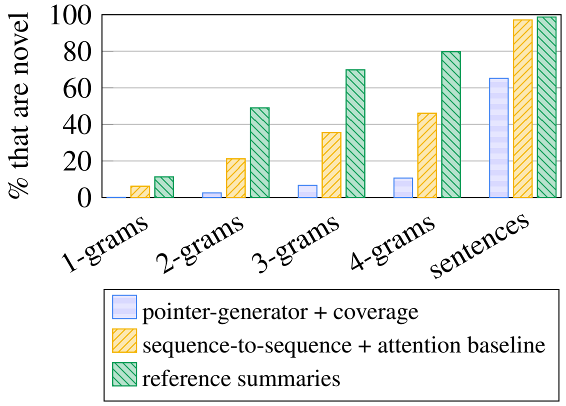

Figure 6: Although our **best model** is abstractive, it does not produce novel n-grams (i.e., n-grams that don't appear in the source text) as often as the **reference summaries**. The **baseline model** produces more novel n-grams, but many of these are erroneous (see section 7.2).

Article: andy murray (...) is into the semi-finals of the miami open , but not before getting a scare from 21 year-old austrian dominic thiem, who pushed him to 4-4 in the second set before going down 3-6 6-4, 6-1 in an hour and three quarters. (...)
Summary: andy murray **defeated** dominic thiem 3-6 6-4, 6-1 in an hour and three quarters.

Article: (...) wayne rooney smashes home during manchester united 's 3-1 win over aston villa on saturday. (...)
Summary: manchester united **beat** aston villa 3-1 at old trafford on saturday.

Figure 7: Examples of abstractive summaries produced by our model (**bold** denotes novel words).

In particular, Figure 6 shows that our final model copies whole article sentences 35% of the time; by comparison the reference summaries do so only 1.3% of the time. This is a main area for improvement, as we would like our model to move beyond simple sentence extraction. However, we observe that the other 65% encompasses a range of abstractive techniques. Article sentences are truncated to form grammatically-correct shorter versions, and new sentences are composed by stitching together fragments. Unnecessary interjections, clauses and parenthesized phrases are sometimes omitted from copied passages. Some of these abilities are demonstrated in Figure 1, and the supplementary material contains more examples.

Figure 7 shows two examples of more impressive abstraction – both with similar structure. The dataset contains many sports stories whose summaries follow the *X beat Y* $\langle$*score*$\rangle$ *on* $\langle$*day*$\rangle$ template, which may explain why our model is most confidently abstractive on these examples. In general however, our model does not routinely produce summaries like those in Figure 7, and is not close to producing summaries like in Figure 5.

The value of the generation probability p_{gen} also gives a measure of the abstractiveness of our model. During training, p_{gen} starts with a value of about 0.30 then increases, converging to about 0.53 by the end of training. This indicates that the model first learns to mostly copy, then learns to generate about half the time. However at test time, p_{gen} is heavily skewed towards copying, with a mean value of 0.17. The disparity is likely due to the fact that during training, the model receives word-by-word supervision in the form of the reference summary, but at test time it does not. Nonetheless, the generator module is useful even when the model is copying. We find that p_{gen} is highest at times of uncertainty such as the beginning of sentences, the join between stitched-together fragments, and when producing periods that truncate a copied sentence. Our mixture model allows the network to copy while simultaneously consulting the language model – enabling operations like stitching and truncation to be performed with grammaticality. In any case, encouraging the pointer-generator model to write more abstractively, while retaining the accuracy advantages of the pointer module, is an exciting direction for future work.

8 Conclusion

In this work we presented a hybrid pointer-generator architecture with coverage, and showed that it reduces inaccuracies and repetition. We applied our model to a new and challenging long-text dataset, and significantly outperformed the abstractive state-of-the-art result. Our model exhibits many abstractive abilities, but attaining higher levels of abstraction remains an open research question.

9 Acknowledgment

We thank the ACL reviewers for their helpful comments. This work was begun while the first author was an intern at Google Brain and continued at Stanford. Stanford University gratefully acknowledges the support of the DARPA DEFT Program AFRL contract no. FA8750-13-2-0040. Any opinions in this material are those of the authors alone.